Speech Emotion Recognition

DATA 255 Deep Learning

Group 2

Meet the team

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Contents

1. Project Domain/Problem

Domain: **Speech Emotion Recognition (SER)** automatically detect human emotional states from **spoken audio**.

Application

- Mental health monitoring
- Automotive safety (driver stress detection)
- Customer service analytics
- Human-robot interaction

Core Challenges

Static Feature Modeling	Class Imbalance	Speaker Variability	Cross-corpus generalization
Most CNN-based models treat audio as a static image	models default to predicting majority classes	Learn speaker identity instead of emotional content	Trained on clean lab data
Cannot capture how emotion evolves over time	**Neutral and sad are underrepresented**	**Same emotion sounds different across speakers**	**Fail in real-world noisy environments**

Beyond the Model: Real-World Impact

Live Call Center Escalation Prevention

- **The Problem:** Agents often only realize a customer is upset when they start yelling, making it too late to de-escalate or save the relationship.
- **The SER Solution:** Real-time acoustic monitoring tracks the conversational *trajectory*, detecting subtle shifts from "Neutral" to "Frustrated."
- **The Value:** Proactively flashes de-escalation tips on the agent's screen or silently routes the call to a manager *before* the situation explodes.

Telehealth & Mental Wellness Tracking

- **The Problem:** Over virtual microphones, it is notoriously difficult for clinicians to distinguish between a patient having a "Neutral" (flat) affect versus genuine "Sadness."
- **The SER Solution:** Analyzes objective, underlying audio frequencies to isolate subtle emotional cues that the human ear might miss over a digital connection.
- **The Value:** Generates a data-driven "emotional timeline" of the session, giving therapists an objective tool to track depressive symptoms and progress over time.

2 Baseline – Technical Baseline

Existing Model	Architecture	Limitation
MFCC + MLP (Badshah et al. 2019)	Extracts handcrafted MFCC features → fully connected layers	No spatial or temporal modeling, treats entire utterance as a static vector, ignores how emotion evolves over time
Transformer (Wav2Vec 2.0) (Baevski et al. 2020)	Self-attention architecture pre-trained on 960 hrs of unlabeled audio	Computationally too heavy for real-time deployment, completely dependent on massive pre training data, without it, performance drops to ~45%

Remaining gaps across existing works

- No modeling of emotional continuity across utterances in a conversation
- Class imbalance unaddressed, neutral consistently ignored
- Speaker variability not handled, models learn speaker identity over emotion
- Accuracy vs real-time deployment trade-off unsolved
- Poor generalization from clean lab data to real-world noisy environments

3.1 Datasets & Metric used in Prior Work

Model	Dataset	Accuracy
CNN (Badshah 2019)	IEMOCAP	~60% (with pretraining)
wav2vec 2.0 (Baevski 2020)	IEMOCAP	~70% (960hrs pretraining)

Why choose these two models as baseline?

MFCC + MLP *(Badshah et al. 2019)*

- Represents the simplest possible approach, handcrafted features, no spatial or temporal modeling
- Serves as lower bound, establishes minimum performance any meaningful model must exceed
- Classic SER baseline, widely used in literature for fair comparison

Transformer / wav2vec 2.0 *(Baevski et al. 2020)*

- Represents current SOTA architecture in SER
- Serves as upper bound reference, 70%+ with full pretraining
- Tests whether architectural complexity alone is sufficient without pretraining
- Wagner et al. (2023) identified it as too heavy for real-time deployment,we verify this claim

3.2 Datasets & Metric in Our Model

We use a staged dataset strategy instead of mixing all datasets at once. This gives us a cleaner benchmark, avoids early label and domain mismatch, and better supports the professor's feedback to model temporal emotion change rather than only static clip-level classification like RAVDESS.

IEMOCAP

- 12 hours of dyadic conversational speech
- 10 speakers (5 male, 5 female), 5 sessions
- 4-class subset: angry, happy, sad, neutral
- Industry standard benchmark for SER

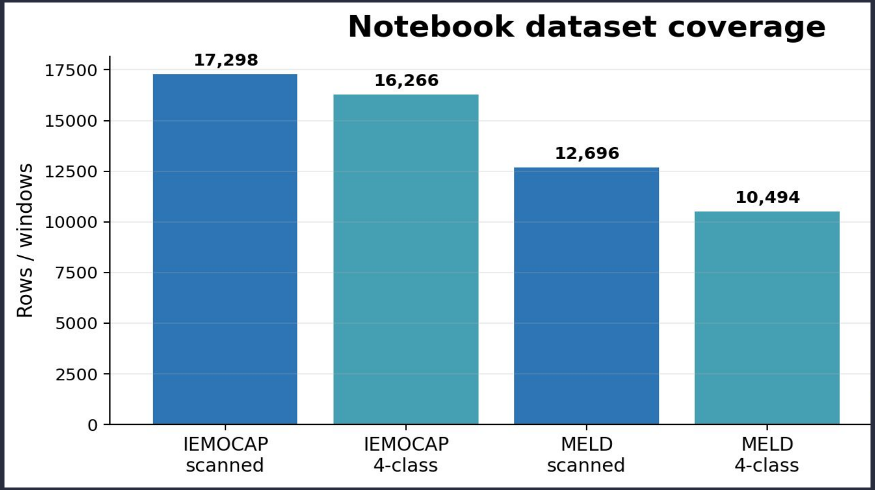
MELD

- Extracted from Friends TV show
- 13,708 utterances, 7 emotion classes
- Real-world noise: background music, laughter, reverberation
- Used in v3 as noisy out-of-domain training data

Metric	Description
Accuracy	Overall correct predictions
Macro-F1	Equal weight per class, handles imbalance
WAR	Weighted Average Recall
UAR	Unweighted Average Recall — SER standard
RTF	Inference time / audio duration — deployment

3.3 Prerequisites: Data Readiness

Notebook EDA, feature extraction, and loader checks before modeling.



WHAT THE NOTEBOOK ESTABLISHES

- **IEMOCAP** is the main speech-emotion classification source.
- **MELD** is used for dialogue context and external generalization checks.
- All preprocessing cells are modular: extraction, EDA, label mapping, loaders.

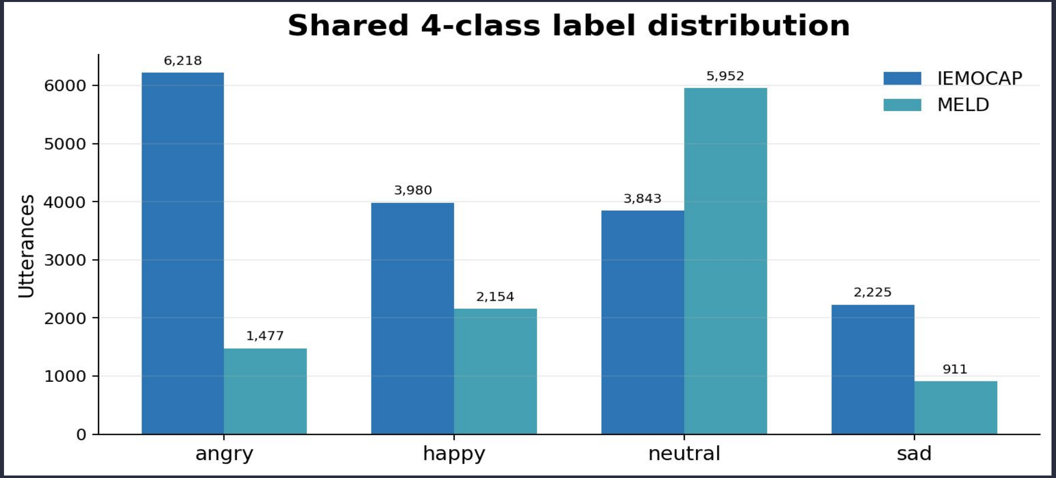
KEY PREREQUISITES CONFIRMED

16,266
IEMOCAP shared-label rows

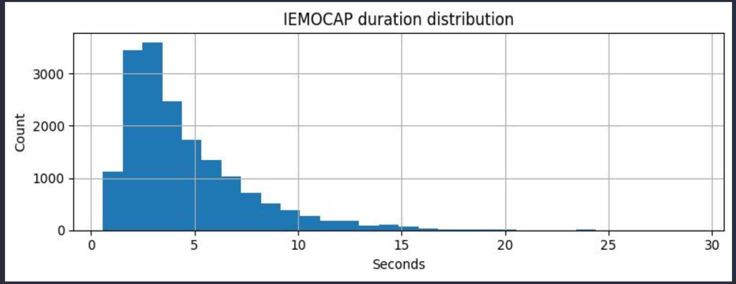
10,494
MELD shared-label rows

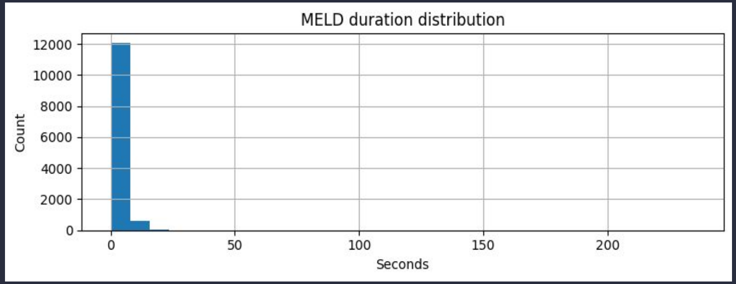
3.4 EDA Snapshot: Labels & Duration

Label balance and audio length checks before feature extraction.



DURATION CHECKS FROM EDA





EDA TAKEAWAYS

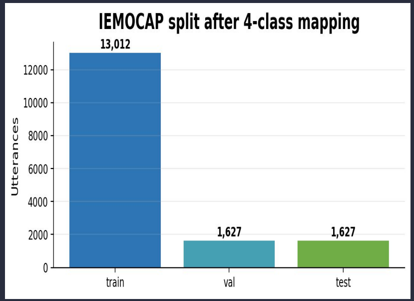
- Shared label space keeps four target classes: angry, happy, sad, neutral.
- IEMOCAP becomes moderately skewed after merging frustration → angry and excited/happiness → happy.
- MELD is neutral-heavy, so Macro-F1 and confusion analysis remain important.
- Duration histograms justify fixed-length padding/truncation before batched training.

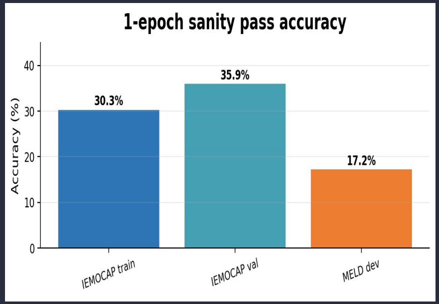
3.5 Preprocessing & Loader Checks

Audio is converted into a consistent feature image before model training.

FEATURE CONFIGURATION

- TARGET_SR = 16,000 Hz, MAX_SECONDS = 6.0, MAX_LEN = 96,000 samples.
- Log-Mel uses 64 mel bands; MFCC uses 20 coefficients + delta + delta-delta.
- FEATURE_MODE = both → concatenated 124 × 376 feature image per utterance.
- DataLoader emits hard labels and one-hot soft targets for each row.





LOADER/READINESS CHECKS

- IEMOCAP split: 13,012 train / 1,627 validation / 1,627 test rows.
- Quick-run subset validates the pipeline before longer training jobs.
- MELD dev sanity score highlights expected cross-corpus generalization gap.

4 Our Approach vs Baseline Models

Technical novelty — addressing literature gaps

Identified limitation	v2 solution	v3 addition
CNNs treat audio as static image No memory of how emotion evolves over time within an utterance Badshah 2019 · Zhang 2019	**Bi-LSTM + Attention** models temporal dynamics in both directions	**Same** (retained)
LSTMs lose emotional cues in long speech — each utterance treated independently, no conversational history · Li 2022	**Dialogue context** prev 2 utterances fed as input to track emotion shifts	**Extended to** prev 3 utterances for richer emotional history
Models overfit majority classes Neutral and sad ignored during training due to class imbalance Schuller 2021 · Latif 2021	**Focal Loss +** class weights + label smoothing + augmentation	**+ Mixup** augmentation interpolates minority samples
Speaker variability confounds models learn speaker identity instead of emotion features Latif 2021 · Zhao 2021	**BatchNorm** reduces inter-speaker variance (partial)	**InstanceNorm2d** normalizes per sample — removes speaker tone bias
SOTA models too heavy wav2vec 2.0 cannot deploy on edge devices in real-time Wagner 2023 · Baevski 2020	**Lightweight** dual-branch CNN no pretraining RTF = 0.00008	**Same** RTF = 0.00008 (retained)

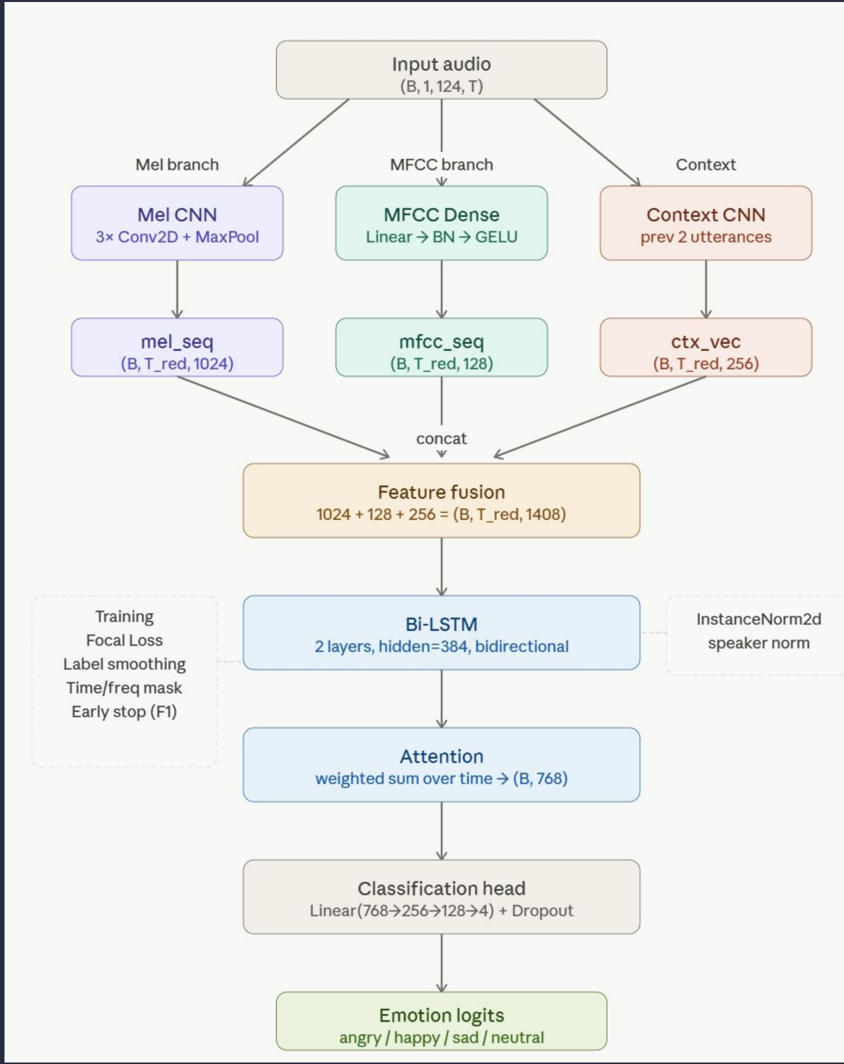
Dialogue Context takes previous 2-3 utterances as input, capturing emotional shifts across a conversation

Focal Loss + class weights + label smoothing neutral F1 improved, resolving prediction failure on minority classes

InstanceNorm2d (v3) per-sample normalization explicitly removes speaker-specific tone and pitch, forcing model to focus on emotion rather than speaker identity

Lightweight Dual-Branch CNN no pretraining required, fully real-time capable

Cross-domain experiment with MELD (v3) introduced noisy out-of-domain data to test generalization



Feature Extraction

- **Branch 1: 2D CNN on Log-Mel Spectrogram → captures spatial/spectral textures**
- **Branch 2: Dense layers on MFCC + deltas → captures fine-grained temporal coefficients**
- **All branches fused before sequence modeling**

Dialogue-Level Context Modeling

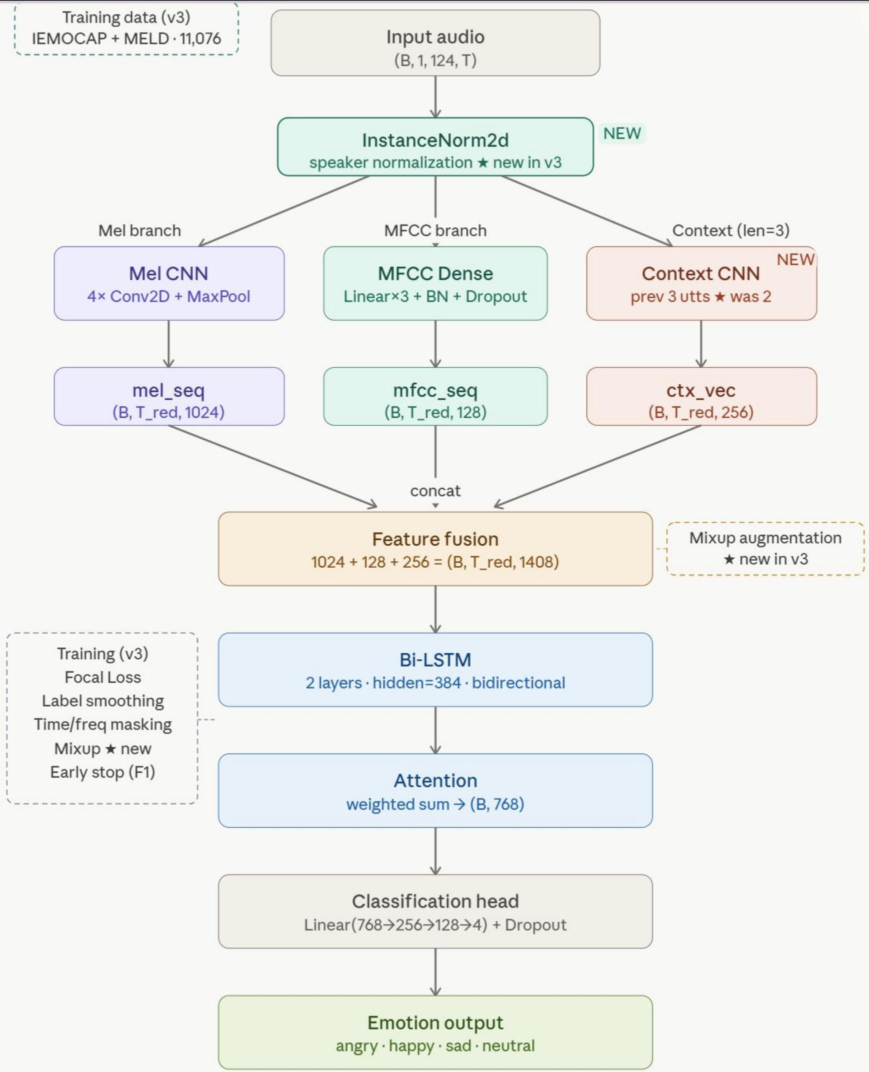
- **A dedicated Context CNN encodes the previous 2 utterances, capturing conversational emotional history**
- **Captures how emotion evolves across a conversation, not just a single utterance**

Bi-LSTM + Attention

- **Bidirectional LSTM models temporal dynamics in both directions**
- **Attention mechanism focuses on the most emotionally salient time steps**

Training

- **Focal Loss to address class imbalance (neutral was completely ignored before)**
- **Label Smoothing to prevent overconfidence**
- **Data augmentation: time masking, frequency masking, Gaussian noise**



Model V3

1. **Speaker Normalization (InstanceNorm2d)**

2. **Mixup augmentation**

3. **Extended context (3 prev utterances instead of 2)**

4. **Introduce Noisy and extend dataset:** IEMOCAP + MELD combined training data

5.1 Result

V2 (Best Model)

- Trained on: IEMOCAP only (2,831 samples)
- Clean lab recordings
- Results: Accuracy 55.9%, Macro-F1 52.4%, neutral F1 0.38, RTF 0.00008

	precision	recall	f1-score	support
angry	0.67	0.65	0.66	141
happy	0.61	0.59	0.60	100
sad	0.48	0.42	0.45	38
neutral	0.36	0.41	0.38	75
accuracy			0.56	354
macro avg	0.53	0.52	0.52	354
weighted avg	0.57	0.56	0.56	354

V3 (Noisy Dataset Experimental)

- IEMOCAP + MELD data (11,076 samples)
- Improve model robustness and generalization by introducing TV show dialogue with **background noise**.
- Consistent with real-world deployment conditions
- Result drops: Macro-F1 dropped 52.4% → 50.5%

	precision	recall	f1-score	support
angry	0.55	0.74	0.63	141
happy	0.58	0.49	0.53	100
sad	0.55	0.47	0.51	38
neutral	0.47	0.28	0.35	75
accuracy			0.55	354
macro avg	0.53	0.50	0.50	354
weighted avg	0.54	0.55	0.53	354

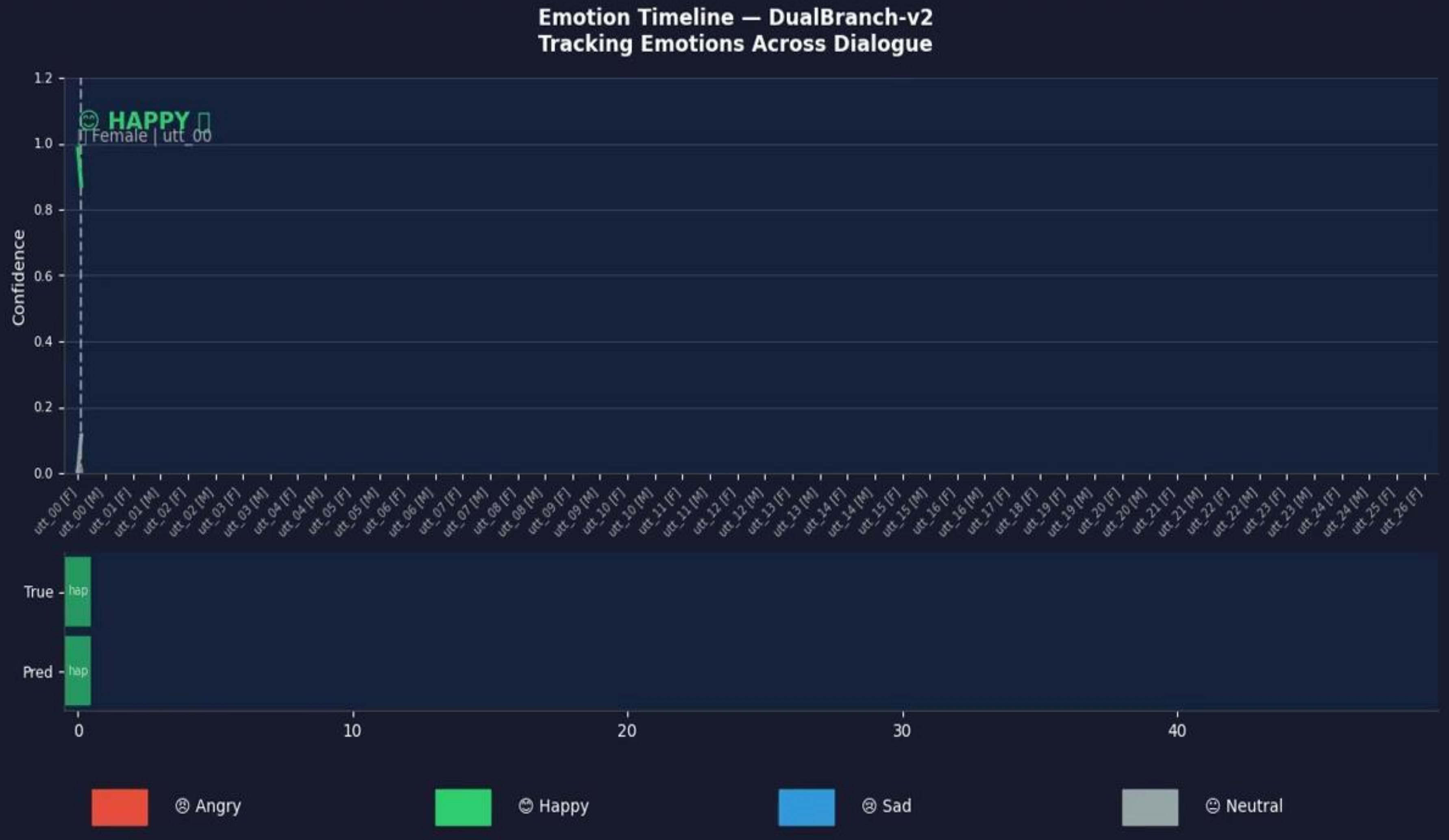
5.1 Result

Model (No Pretraining Weight)	Training Dataset	Accuracy	Macro-F1	UAR	RTF
MFCC + MLP (Baseline)	IEMOCAP	40.1%	38.6%	40.1%	0.00013
Transformer / wav2vec (Baseline)	IEMOCAP	45.5%	43.7%	44.2%	0.00005
DualBranch-v2 (Ours)	IEMOCAP	**55.9%**	**52.4%**	**51.9%**	**0.00008**
DualBranch-v3 (Ours)	IEMOCAP+MELD	54.5%	50.5%	49.7%	0.00008

Key Findings

- **+15.8% Accuracy** over the simplest baseline (MFCC+MLP), demonstrating the benefit of dual-branch spatial-temporal feature extraction combined with dialogue context modeling
- **+7.8% Macro-F1** over the original model, most critically, neutral class F1 improved to 0.38, resolving a prediction failure caused by class imbalance
- **All models achieve real-time performance** (RTF < 0.001), confirming our architecture is suitable for deployment.
- The Transformer baseline confirms Wagner et al. (2023)'s finding: **without pretraining, attention-based models underperform on small datasets**

Demo



Emotion Timeline — DualBranch-v2
Tracking Emotions Across Dialogue
Confidence
😊 HAPPY
Female | utt_00
True
Pred
hap
hap
😠 Angry
😊 Happy
😢 Sad
😐 Neutral

5.2 Future Work

- **Dynamic Speaker-Aware Context**

  Upgrade the fixed 2-utterance context window using **Graph Neural Networks (GNNs)** or Cross-Attention to map the broader emotional flow between specific speakers.

- **Supervised Contrastive Learning**

  Implement contrastive loss to explicitly maximize the latent space boundary between highly confused classes.

- **Real-Time Edge Deployment**

  Leverage the architecture's highly optimized Real-Time Factor (RTF: 0.00008) by exporting to ONNX/TensorRT for live, low-latency applications

- **Foundation Model Integration**

  incorporate pre-trained weights from large-scale models like wav2vec 2.0. Since our research confirmed these models rely on massive pre-training to achieve SOTA results, leveraging their pre-trained acoustic features will drastically elevate our model's baseline accuracy.

Appendix Team Contribution

Abhinita | Data & Feature Engineering

- Led the initial pipeline setup, dataset acquisition, and data cleaning.
- Managed cross-dataset label coordination to ensure consistent emotional categories.
- Engineered the core acoustic feature extraction (Log-Mel Spectrograms and MFCCs).

Arya | Core Branch Architecture

- Processed and aligned the extracted audio features for model ingestion (shape matching).
- Designed and implemented the foundation of the dual-branch architecture.
- Developed the specific Spatial CNN and Temporal extraction blocks.

Anshika | Sequence Modeling & Fusion

- Architected the feature fusion pipeline to combine the dual-branch outputs.
- Integrated the **Bi-LSTM** network to capture the temporal flow of the audio.
- Implemented the **Attention block** and final Fully Connected (FC) classification layers.

Jane | Optimization & Contextual Logic

- Spearheaded model tuning to elevate metrics for real-world deployment readiness.
- Tackled dataset imbalance by introducing **Focal Loss** and data augmentation techniques.
- Engineered the multi-utterance **Dialog Context** module and integrated InstanceNorm2d for training stability.

Thank you